

solarArrayReference

1 Executive Summary

This module computes the reference rotation angle θ_R for a single-axis solar array drive. In the default `AUTO_TRACK` mode, θ_R is the angle that aligns the solar array surface normal with the Sun direction as well as possible (perfect incidence is achievable when the drive axis and the Sun direction are perpendicular). In `SPECIFIED_ANGLE` mode, the module ignores the Sun direction and outputs a user-supplied fixed angle. In both modes, an optional offset angle is added before the result is wrapped to $[-\pi, \pi]$.

2 Message Connection Descriptions

The following table lists all the module input and output messages. The module msg connection is set by the user from python. The msg type contains a link to the message structure definition, while the description provides information on what this message is used for.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
attNavInMsg	NavAttMsgF32Payload	input attitude navigation message containing $\sigma_{B/N}$ and the Sun direction in body-frame components
attRefInMsg	AttRefMsgF32Payload	input attitude reference message containing $\sigma_{R/N}$
hingedRigidBodyInMsg	HingedRigidBodyMsgF32Payload	input hinged rigid body message containing the current panel angle θ_C
hingedRigidBodyRefOutMsg	HingedRigidBodyMsgF32Payload	output hinged rigid body reference message containing the reference angle θ_R

3 Module Parameters

The following table lists all the module parameters than can be set. The parameters are optional unless indicated (if not specified default is used).

Table 2: Module Parameters

Parameter Name	Type	Units	Default	Description	Bounds
driveAxis (required)	Eigen::Vector3f [-]		zero	Solar array drive axis ${}^B\hat{\mathbf{a}}_1$ in body frame, set via <code>setSolarArrayAxes.B</code>	Norm must be within <code>1e-3</code> of 1.0 (checked in setter; renormalized on storage)

Parameter Name	Type	Units	Default	Description	Bounds
surfaceNormal (required)	Eigen::Vector3f [-]		zero	Solar array surface normal ${}^B\hat{\mathbf{a}}_2$ at zero rotation, set via <code>setSolarArrayAxes_BdriveAxis</code>	Norm must be within 1e-3 of 1.0 and orthogonal to <code>driveAxis</code> (absolute value of dot product less than 1e-5 after normalization); re-orthogonalized against <code>driveAxis</code> on storage
alignmentThreshold	float	[rad]	1e-3	Threshold angle ϵ_a between the Sun direction and the drive axis below which the Sun is considered aligned with the drive axis (no preferred rotation)	Must be in $[0, \pi/2]$ (checked in setter)
trackingMode	TrackingMode [-]		AUTO_TRACK	Selects between Sun-tracking (AUTO_TRACK) and fixed-angle (SPECIFIED_ANGLE) reference computation	N/A
specifiedArrayAngle	float	[rad]	0	Reference angle returned when <code>trackingMode</code> is SPECIFIED_ANGLE	Any value accepted (wrapped to $[-\pi, \pi]$ after the offset is applied)
offsetAngle	float	[rad]	0	Offset added to the computed reference angle before wrapping	Any value accepted (wrapped to $[-\pi, \pi]$ together with the base reference)

4 Module Assumptions and Limitations

This module computes the rotation angle required to achieve the best incidence angle between the Sun direction and the solar array surface. This does not mean that perfect incidence (Sun direction perpendicular to array surface) is guaranteed: perfect incidence is only achievable when the drive axis and the Sun direction are perpendicular. Conversely, when they are parallel, no power generation is possible, and in that case the reference is set to the current panel angle to avoid pointless rotation.

The drive axis $\hat{\mathbf{a}}_1$ and surface normal $\hat{\mathbf{a}}_2$ are assumed to be fixed in the body frame. The Sun direction in body-frame components is extracted from `attNavInMsg` and is mapped into the reference frame using the body and reference attitudes from `attNavInMsg` and `attRefInMsg`, so the reference angle is computed in the frame that the spacecraft will occupy at the end of the active slew.

The output reference angle θ_R is always wrapped to $[-\pi, \pi]$.

5 Initialization

The module is configured by:

```
module = solarArrayReferenceF32.SolarArrayReference()
module.modelTag = "solarArrayReference"
module.setSolarArrayAxes_B([1.0, 0.0, 0.0], [0.0, 1.0, 0.0])
module.alignmentThreshold = 1e-3
module.offsetAngle = 0.0
```

For AUTO_TRACK mode (the default), no further configuration is required. To use SPECIFIED_ANGLE mode:

```
module.trackingMode = solarArrayReferenceF32.TrackingMode_SPECIFIED_ANGLE
module.specifiedArrayAngle = 0.5
```

6 Detailed Module Description

For this module to operate, the user provides two body-fixed unit directions:

- ${}^B\hat{\mathbf{a}}_1$: drive axis, about which the array rotates;
- ${}^B\hat{\mathbf{a}}_2$: surface normal at zero rotation.

These vectors must be (near-)unit length and orthogonal. The setter `setSolarArrayAxes_B` validates the norms, verifies orthogonality, normalizes both inputs, and re-orthogonalizes $\hat{\mathbf{a}}_2$ against $\hat{\mathbf{a}}_1$ for exact orthogonality on storage.

6.1 Algorithm Flow

At every update cycle, the `solarArrayReference` module performs the following steps:

1. **Select tracking mode:**

- If `trackingMode` is AUTO_TRACK, compute θ_R from the Sun direction (steps 2-4).
- If `trackingMode` is SPECIFIED_ANGLE, set $\theta_R = \theta_{\text{specified}}$ and skip to step 5.

2. **Map Sun direction to reference frame:** normalize the body-frame Sun direction ${}^{B_C}\hat{\mathbf{r}}_S$ (using `stableNormalized`), then map it into the reference frame using the DCM $[\mathcal{R}\mathcal{B}] = [\mathcal{R}\mathcal{N}][\mathcal{B}\mathcal{N}]^T$:

$${}^{\mathcal{R}}\hat{\mathbf{r}}_S = [\mathcal{R}\mathcal{B}] {}^{B_C}\hat{\mathbf{r}}_S$$

3. **Check Sun/drive-axis alignment:** compute the angle between the Sun direction and the drive axis,

$$\alpha = \arccos(|{}^{\mathcal{R}}\hat{\mathbf{r}}_S \cdot \hat{\mathbf{a}}_1|)$$

If $\alpha < \epsilon_a$ (or the Sun direction is the zero vector), the Sun is aligned with the drive axis and there is no preferred rotation: set $\theta_R = \theta_C$ (the current panel angle from `hingedRigidBodyInMsg`) and skip to step 5.

4. **Compute reference angle** (Sun not aligned with drive axis):

- Project the Sun direction onto the rotation plane and normalize:

$${}^{\mathcal{R}}\hat{\mathbf{a}}_2^{\text{ref}} = \frac{{}^{\mathcal{R}}\hat{\mathbf{r}}_S - ({}^{\mathcal{R}}\hat{\mathbf{r}}_S \cdot \hat{\mathbf{a}}_1) \hat{\mathbf{a}}_1}{|{}^{\mathcal{R}}\hat{\mathbf{r}}_S - ({}^{\mathcal{R}}\hat{\mathbf{r}}_S \cdot \hat{\mathbf{a}}_1) \hat{\mathbf{a}}_1|}$$

- The reference angle is the signed angle between $\hat{\mathbf{a}}_2$ and ${}^{\mathcal{R}}\hat{\mathbf{a}}_2^{\text{ref}}$:

$$\theta_R = \pm \arccos(\hat{\mathbf{a}}_2 \cdot {}^{\mathcal{R}}\hat{\mathbf{a}}_2^{\text{ref}})$$

The sign is determined by the orientation of $\hat{\mathbf{a}}_2 \times {}^{\mathcal{R}}\hat{\mathbf{a}}_2^{\text{ref}}$ relative to the drive axis $\hat{\mathbf{a}}_1$: if the cross product is anti-parallel to $\hat{\mathbf{a}}_1$, the sign is negative.

5. **Apply offset and wrap:** add the configured offset angle and wrap the result to $[-\pi, \pi]$:

$$\theta_R \leftarrow \text{atan2}(\sin(\theta_R + \theta_{\text{offset}}), \cos(\theta_R + \theta_{\text{offset}}))$$

6.2 Wrapping and the Sun-Aligned Case

The arccos-based computation in step 4 produces values in $[0, \pi]$; the sign correction extends this to $[-\pi, \pi]$. Adding the offset can push the sum outside this range, so the final $\text{atan2}(\sin(\cdot), \cos(\cdot))$ step is what guarantees the output range is always $[-\pi, \pi]$.

When the Sun direction is (nearly) aligned with the drive axis, the projection in step 4 is degenerate. The module detects this via the `alignmentThreshold` check and falls back to returning the current panel angle θ_C plus the offset. Because the wrap is applied unconditionally at the end, this also normalizes θ_C (which the caller may have provided unwrapped) to $[-\pi, \pi]$.

6.3 Specified Angle Mode

In `SPECIFIED_ANGLE` mode the module bypasses the Sun-tracking computation entirely and returns the user-supplied `specifiedArrayAngle`. The offset angle and the final $[-\pi, \pi]$ wrap are still applied, so:

$$\theta_R = \text{atan2}(\sin(\theta_{\text{specified}} + \theta_{\text{offset}}), \cos(\theta_{\text{specified}} + \theta_{\text{offset}}))$$

This mode is useful for parking the array at a known orientation (e.g. during slews or eclipse) without depending on a valid Sun direction measurement.